FINAL YEAR CS PROJECT – POC PRESENTATION

Bit-By-Bit – Proof of Concept

Project Introduction

GameStart is an online e-commerce platform designed for gamers to explore and purchase video games, consoles, and accessories. It works as fully digital version of common brick and mortar game stores, allowing users to register, log in, browse products, add items to their cart, and check out securely.

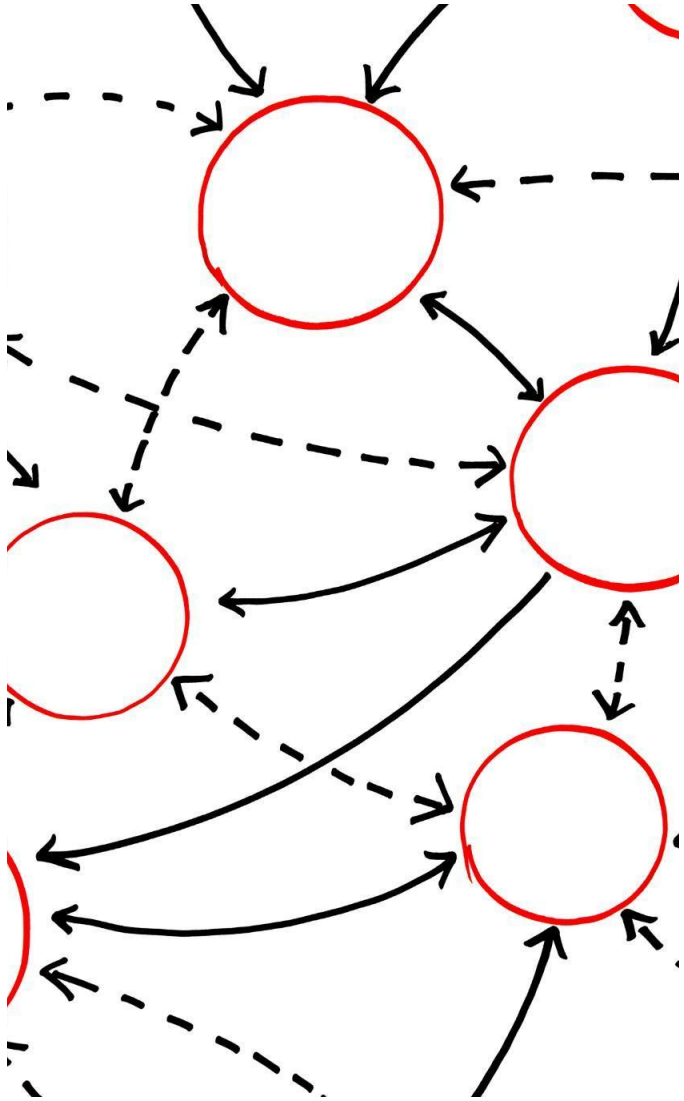
Team Members and Instructor details:

- Fazlul H Faizal: Backend Developer, System Architecture Design
- Moh Prajapati: Document Creator/ Frontend Developer
- Joshua Delshad: Backend, LEADER, Database
- Developer
- Fotios Bampouridis: Frontend Developer
- Eitan Abrishami: Frontend Developer

Course :

- COMP 490/L

Logo: GameStart



Problem Statement

Core Problem Description

- Gamers lack a unified, user-friendly platform to browse, compare, and buy games/consoles with smart recommendations and game data stats

Affected Stakeholders

- Gamers, retailers, and developers are impacted by fragmented platforms and limited product visibility

Pain Points and Gaps

- Limited real-time game data, ratings, and visuals for informed decisions.
- No personalized recommendations tailored to user preferences.
- Users must visit multiple sites for game stats, preferences, reviews, pricing, and availability, creating a fragmented buying experience.

Visual Problem Flow

- Diagram illustrating the transition from current state to identified pain points, enhancing audience understanding.

Objective / Goals

Demonstrate Feasibility

- Deliver a working mobile e-commerce app built specifically for gaming products.
- Integrate live game data using the RAWG API to prove real-time external service connectivity.

Implement Core Functionalities

- Allow users to register/login, browse categories, choose platforms, and add items to the cart.
- Display product details, pricing, and images dynamically inside the Search/Product screens.

Validate User Flows

- Verify seamless navigation across Home → Category Selection → Product Grid → Cart → Checkout.
- Ensure category choices correctly filter and display related products.

Success Criteria Checklist

- Users can browse, add to cart, and reach checkout without errors.
- Core features navigation tabs, cart operations, and RAWG API data function consistently and reliably.

	Feature	Category	Description	Status
1	Bottom tab navigation	Must-have	Navigation tabs for Home, Search, Cart, and Profile	Implemented
2	Core shopping flow	Must-have	Browse products, add to cart, checkout functionality	Implemented
3	Platform selection modal	Nice-to-have	Modal interface for filtering products by gaming platform	Planned
4	Profile management	Nice-to-have	Access to orders, wishlist, settings, and recent history	Planned
5	Category browsing	Must-have	Browse products by categories (games, consoles, accessories)	Implemented
6	Search filters	Must-have	Filter search results by various criteria	Implemented
7	Cart operations	Must-have	Add, remove, and modify items in shopping cart	Implemented
8	Log / sign up	Must-have	Login and account management functionality	In Progress
	Dark mode interface	Nice-to-have	Gaming-inspired dark-themed visual style	Implemented
10	AI recommendations	Nice-to-have	Smart product recommendations based on user preferences	Planned
11	Database	Must-have	User login credential	In Progress

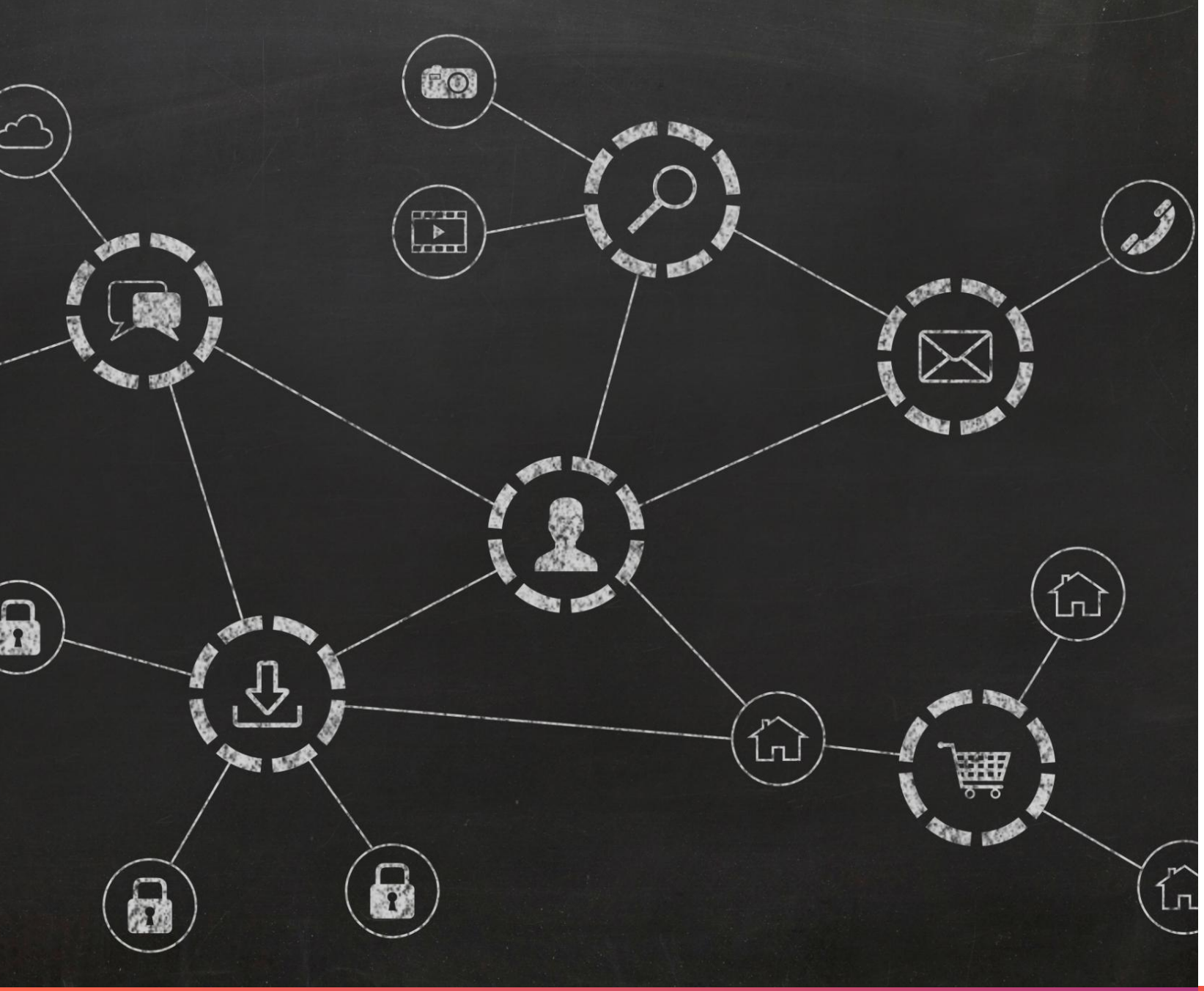
Key Features / Functionality

Must-Have Features

- Bottom tab navigation:
  - Home, Search, Cart, Profile
- Core shopping flow:
  - browse products, add to cart, checkout

Nice-to-Have Features

- Platform selection modal for category filtering
- Profile options: orders, wishlist, settings, recent history



Architecture Overview

System Components

Client/UI: React Native mobile app with tab navigation

Backend Service Layer: Python Django for APIs, authentication, logic, database management

External Data Layer: RAWG API for live game details and statistics

Database/Storage: MongoDB, Stores user login credentials, user orders

Data Flow Representation

User → React Native App, RAWG API → Django Backend → Database → Response Back to App

Arrows show how requests move from the app to the backend and return processed data to the UI.

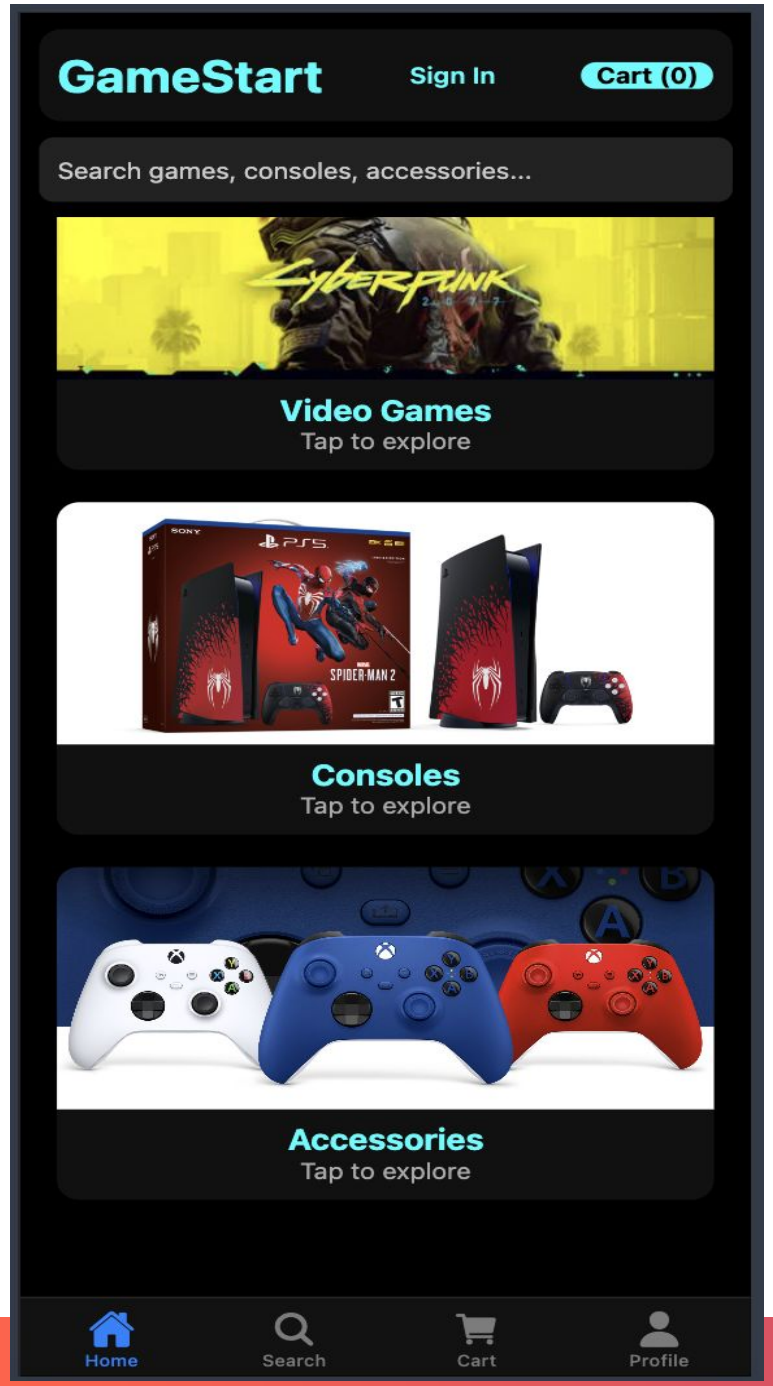
Technology Stack Legend

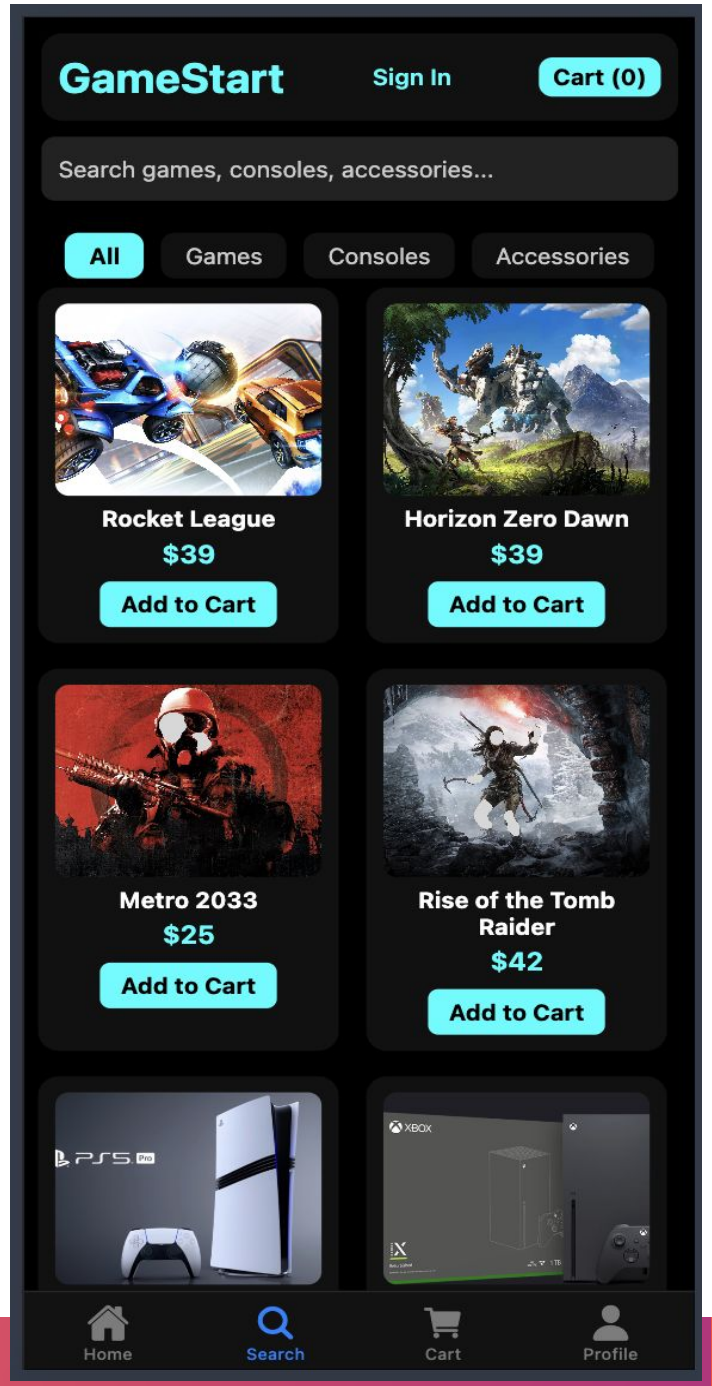
Frontend: React Native

Backend: Python Django

External Services: RAWG API (game catalog + metadata)

Database: MongoDB with dynamic noSQL storage





Design & Mockups

Visual Design Overview

- Displays core app screens (Home, Search, Cart, Profile) to show interface layout and navigation flow.
- Highlights the gaming-inspired dark theme

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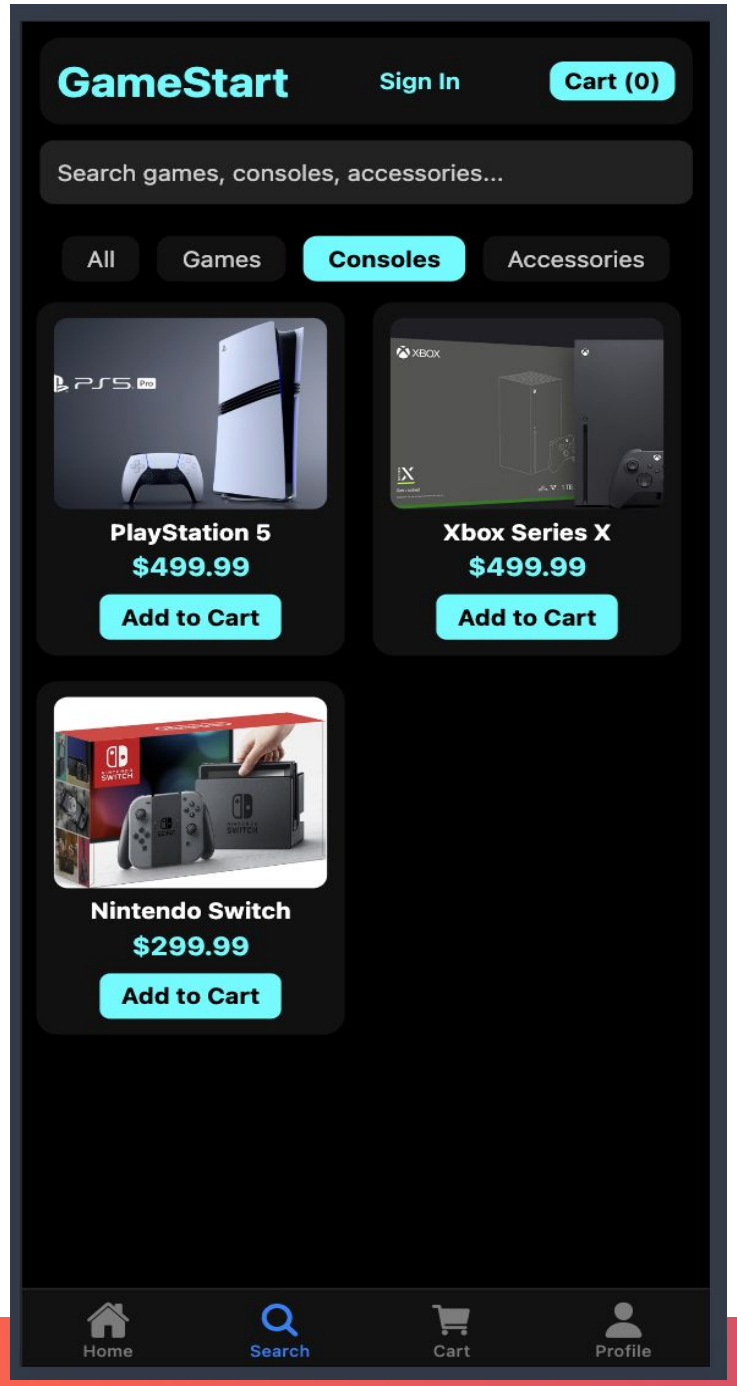
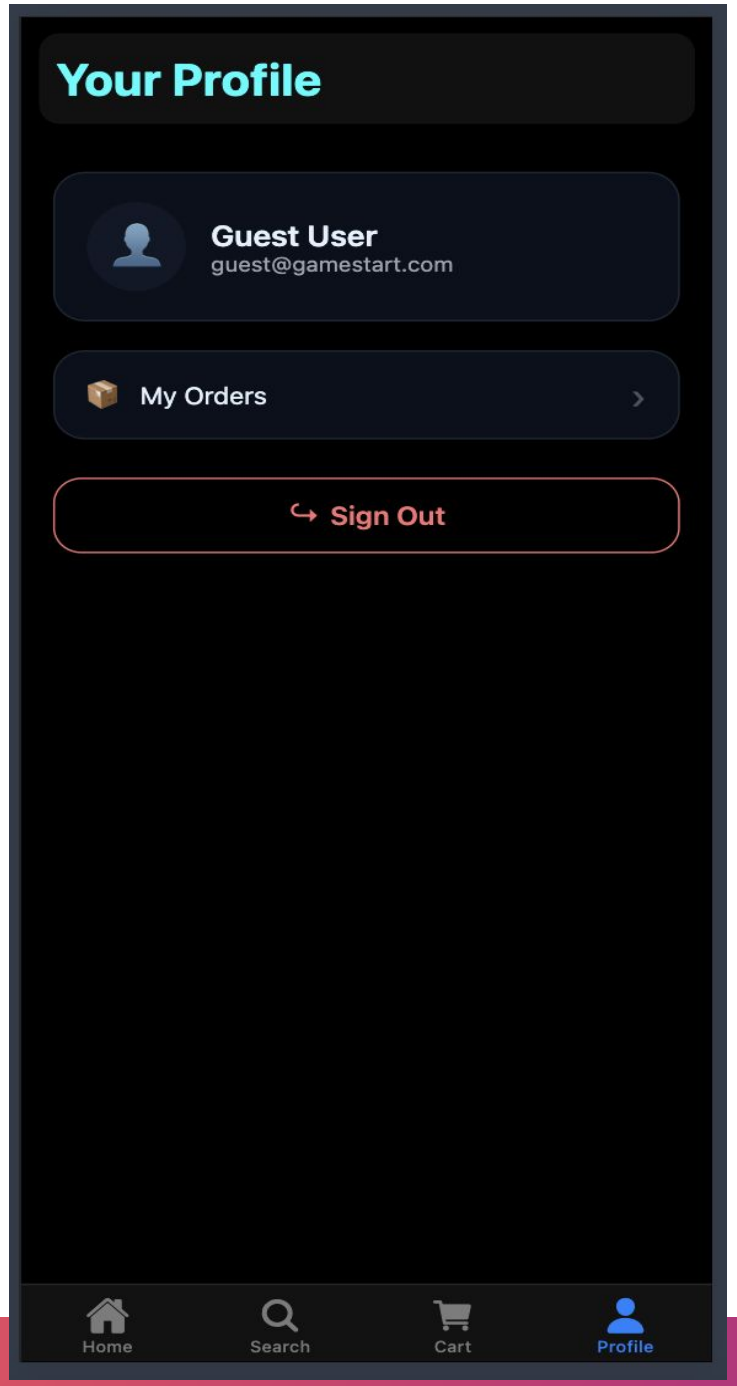
User Experience Considerations

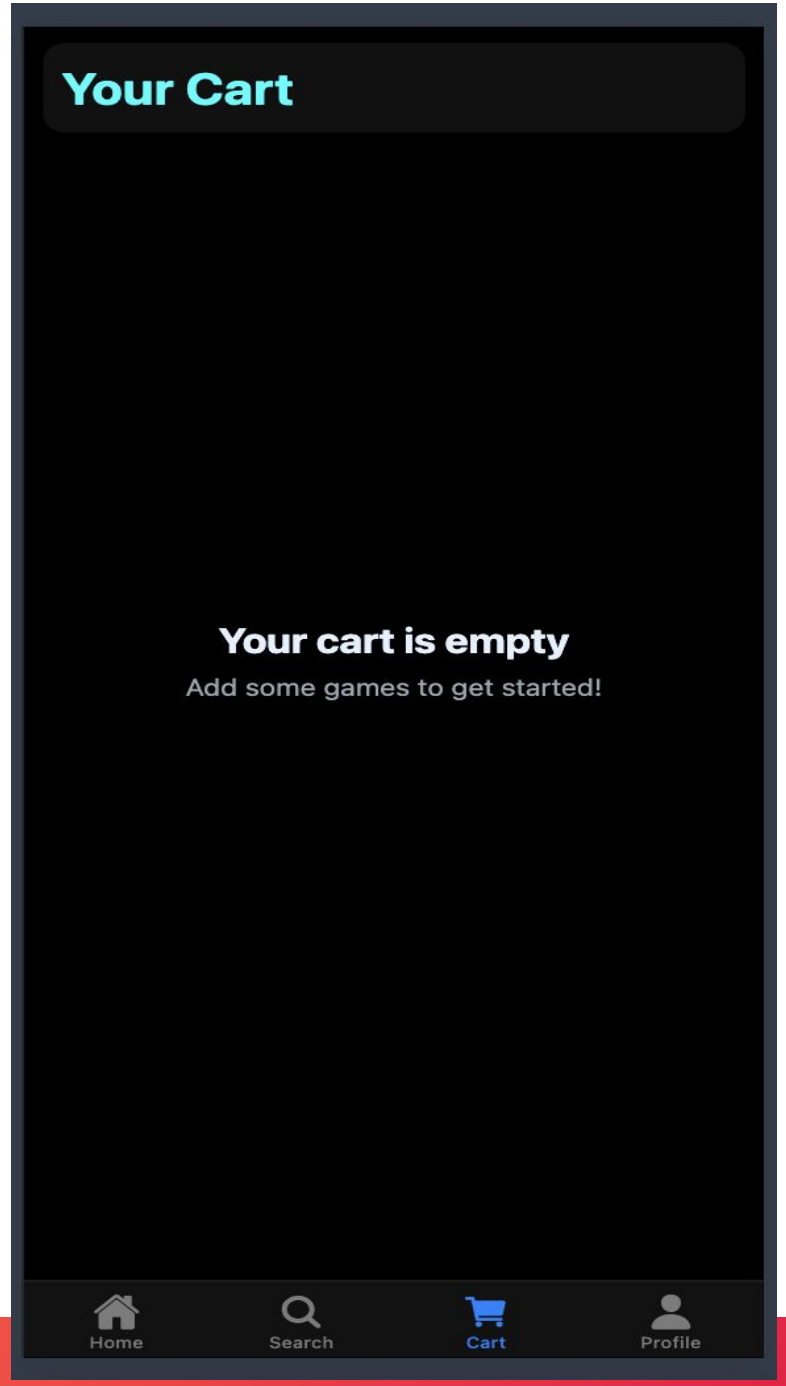
- Designed for minimal clicks and clear navigation using bottom tabs.
- High-contrast colors and large buttons improve readability and touch usability.

User-Centered Design Focus

- Layout and interactions prioritize how gamers explore products and manage carts.
- Screens and actions reflect real user behaviors, making the app intuitive and familiar.


GameStart
Sign In
Cart (0)
Search games, consoles, accessories...
All
Games
Consoles
Accessories
PlayStation 5
$499.99
Add to Cart
Xbox Series X
$499.99
Add to Cart
Nintendo Switch
$299.99
Add to Cart
Home
Search
Cart
Profile


Your Profile
Guest User
guest@gamestart.com
My Orders
Sign Out
Home
Search
Cart
Profile


Your Cart
Your cart is empty
Add some games to get started!
Home
Search
Cart
Profile

Legal, Security & Compliance Considerations

Data Privacy Measures

- Secure user data and limit access.
- Collect only essential information.

Authentication & Authorization

- Require secure login and role-based access.
- Protect sensitive user actions.

Licensing & IP Compliance

- Follow usage rules for RAWG API and game assets.
- Avoid copyright violations with third-party content.

Phase	Objective	Key Tasks / Activities
Phase 1: Foundation Building & Core Functionality Implementation	Establish core infrastructure	Web framework setup, database creation, product categories, basic UI prototype
	Deploy essential features and systems	Login tests, navigation & cart checks, performance metrics, bug tracking & fixes
Phase 2: Core Functionality Implementation	Ensure practical user experiences	Product management, payment system, order delivery, AI setup, full deployment
	Polish user experience and enhance performance	UI polish, speed improvements, bug fixes, added features
	Go live and maintain system health	Live deployment, user feedback, system monitoring, continuous updates

Testing & Validation

Comprehensive Test Cases

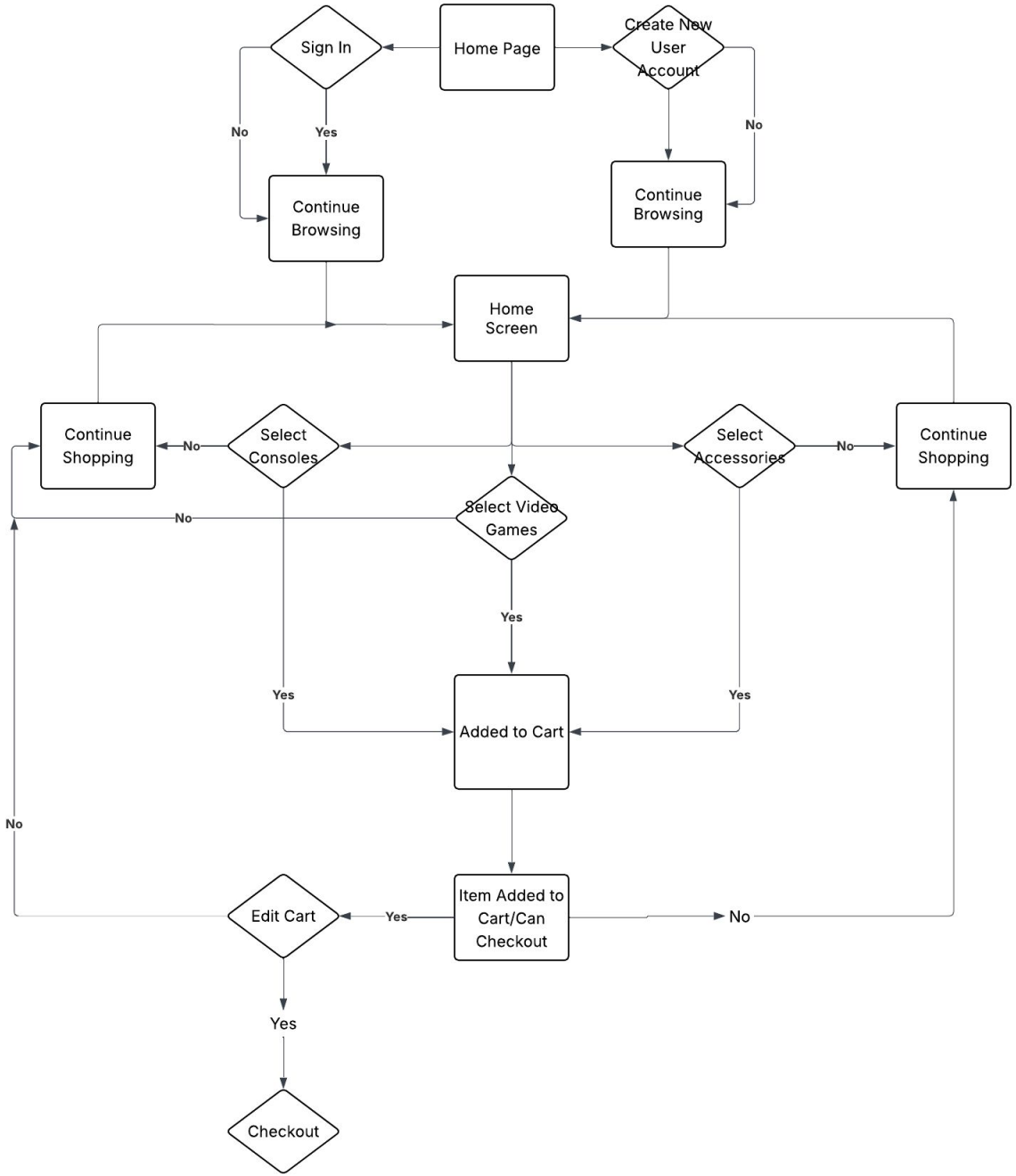
- Test cases cover critical functionalities including login, Validated login, navigation, and cart operations.
- Checked behavior for unauthorized access and screen responsiveness.

Quality Assurance Evidence

- Screenshots and logs confirm expected results and app stability.

Issue Tracking and Limitations

- Documented known bugs and performance gaps for future fixes.



Demo Flow

Setup:

- Open app & show main tabs

Action:

- Select Category
- search
- add to cart

Result:

- View cart with selected items

Feedback & Iteration

Feedback Collection

- Collected input from developers to identify UI and backend improvements.

Addressing UI Clutter

- Simplified layouts and reduced unnecessary elements for clearer navigation.

Optimizing API Performance

- Enhanced API handling for faster load times and smoother interactions.

Visual Before-and-After

- Displayed UI changes to show how feedback improved the design.

Metrics & Achievements

Performance Comparison

- Chart shows improvements in speed, coverage, UX, and stability.

Response Time Improvement

- Faster load times and quicker API responses.

Feature Implementation Rate

- Majority of planned features successfully delivered.

Positive Usability Feedback

- Users found navigation clear and design appealing.

Current and Future Propositions

PHASE 1:

- Core framework
- database web/app integration
- web app search system
- Login/Registration functions
- Web navigation
- Product filtering
- Cart selection
- Profile/Sign-out

PHASE 2:

- Google Oauth
- Payments
- Order processing
- Product planning
- AI Agents for game recommendations
- Full deployment web and database deployment

References / Acknowledgements

Acknowledgements

Instructor: Savita Mittal - Course guidance and project direction

Team Members: *Fazlul H Faizal, Fotios Bampouridis*, *Moh Prajapati, Joshua Delshad, Eitan Abrishami* – Design, development, backend, and integration efforts

Peers & Classmates – Feedback that shaped UI improvements, navigation flow, and usability choices

Transparency and Support

All third-party libraries, and assets were used according to their licensing terms.

Game data and images retrieved from RAWG comply with attribution and usage guidelines.

The team maintained open communication throughout development to ensure accountability, shared ownership, and successful delivery of the proof of concept.